scottm59@msu.edu
mackenzie-scott-121697242
0009-0004-7242-0204
kenzerkay

Mackenzie Ticoras

Education

present **Doctor of Philosophy in Astrophysics and Computational Math, Science and Engineering (dual PhD)**, *Michigan State University*
Advisor: Brian O'Shea

2026 **Master of Science in Physics**, *Michigan State University*

2023 **Bachelor of Science in Physics and Astronomy (double major)**, *The Ohio State University*
Honors Research Distinction and Honors in the College of Arts and Sciences
Undergraduate Thesis: "Refinement of Stellar Intensity Interferometry at VERITAS"
Advisor: Michael Lisa

Scholarships and Fellowships

Mar 2023 National Science Foundation Graduate Research Fellowship
Mar 2023 University Distinguished Fellowship at Michigan State University
Feb 2023 College of Natural Sciences Fellowship at Michigan State University, Decl.
Apr 2022 Undergraduate Research Scholarship at The Ohio State University
Mar 2022 Goldwater Scholarship
May 2021 Physics Summer Research Scholarship at The Ohio State University
2019-2023 Provost Merit Scholarship (Awarded Semesterly)

Awards and Honors

Jun 2026 AAS International Travel Grant
Mar 2023 Physics Senior Award at The Ohio State University
Apr 2022 Best Undergraduate Oral Presentation at EGLS APS Meeting
Apr 2022 Smith Junior Award at The Ohio State University
2019 – 2023 The Ohio State University Dean's List (Awarded 8 Semesters)
2019 – 2023 Undergraduate Honors Program member at The Ohio State University

Successful Computing Proposals

2024 **"Probing Galaxy Formation at Low and High Redshifts"**, *ACCESS Allocation*, 4.3M CPU hours
Role: Co-I

2026 **"Probing Galaxy Formation at Low and High Redshifts (renewal)"**, *ACCESS Allocation*, 2.6M CPU hours
Role: Co-I

Publications

[1] **Mackenzie Ticoras** et al. "Figuring Out Gas & Galaxies in Enzo (FOGGIE). XV. Examining the Spatial and Kinematic Relationship between Circumgalactic Mg ii and O vi". In: *The Astrophysical Journal* 1004.2 (June 2026), p. 159. DOI: 10.3847/1538-4357/ae63ce. URL: https://doi.org/10.3847/1538-4357/ae63ce.

[2] A. Acharyya et al. "An Angular Diameter Measurement of β UMa via Stellar Intensity Interferometry with the VERITAS Observatory". In: *The Astrophysical Journal* 966.1 (Apr. 2024), p. 28. DOI: 10.3847/1538-4357/ad2b68. URL: https://doi.org/10.3847/1538-4357/ad2b68
Corresponding Author: Mackenzie Ticoras.

Research Presentations

Apr 2026 **Milky Way Halos Meeting (Invited)**, *"Examining the Spatial and Kinematic Relationship between Circumgalactic Mg II and O VI"*, virtual

Nov 2024 **Resolving The CGM and Its Impact on Galaxy Evolution**, *"Investigating kinematics and spatial distribution of multiphase gas in the CGM"*, Santa Cruz, Chile

May 2023 **Ohio State Stellar Intensity Interferometry Conference 2023**, *"Measurement of Merak with VERITAS SII"*, Columbus, Ohio

Jan 2023 **VERITAS Collaboration Meeting Winter 2023**, *"An Angular Diameter Measurement of β UMa"*, Tuscon, Arizona

Jul 2022 **OSU Astronomy Department Undergraduate Summer Symposium**, *"A Uniform Disk Visibility Curve for bet UMa"*, Columbus, Ohio

Jun 2022 **240th Meeting of the American Astronomical Society**, *"An intensity interferometry measurement of gamma Orionis (Bellatrix) with the VERITAS telescope array"*, Pasadena, California

Apr 2022 **Eastern Great Lakes Section of the American Physical Society**, *"An intensity interferometry measurement of gamma Orionis (Bellatrix) with the VERITAS telescope array"*, Youngstown, Ohio

Teaching Experience

2025 – pres **Certification in College Teaching (In Progress)**, *Michigan State University*

2021 – 2023 **Student Tutor**, *The Ohio State University*, Math and Statistics Learning Center
Student Tutor 2021; Senior Student Tutor 2021 – 2023

Outreach, Mentorship, and Service

2024 – pres **Outreach Assistant**, *Michigan State University*, Michigan State Observatory
Help facilitate outreach events and public nights at the MSU observatory

2023 – pres **Stellar Mentorship Program**, *Michigan State University*
Mentor 2023 – pres; Program Leader 2024 – pres

2024 – 2025 **Graduate Student Curriculum Committee**, *Michigan State University*

2024 & 2025 **Prospective Student Liason**, *Michigan State University*

Sept 2024 **Michigan Astronomy on Tap**, *"Science and Star Wars"*

2020 – 2023 **Code 4 Community**, *The Ohio State University*
Member 2020 – 2023; Treasurer 2021 – 2022; President 2022 – 2023

Students Mentored

2025 – pres Jazzmin Partirdge